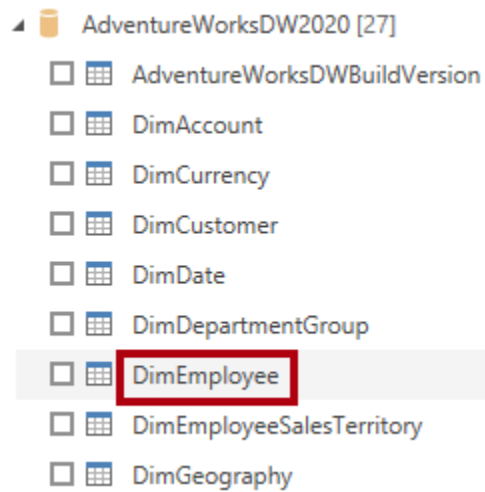


Contents

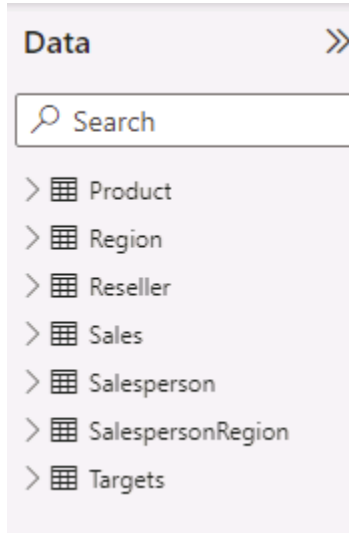
Chapter one	1
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Chapter one

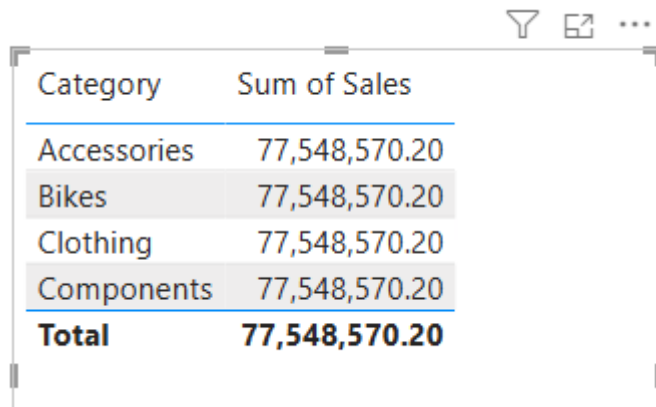
[PL-300-Microsoft-Power-BI-Data-Analyst/Instructions/Labs/01-get-data-in-power-bi.md at Main · MicrosoftLearning/PL-300-Microsoft-Power-BI-Data-Analyst](#)



[PL-300-Microsoft-Power-BI-Data-Analyst/Instructions/Labs/02-transform-data-power-bi.md at Main · MicrosoftLearning/PL-300-Microsoft-Power-BI-Data-Analyst](#)



1. Notice that the table visual lists four product categories, and that the sales value is the same for each, and the same for the total.



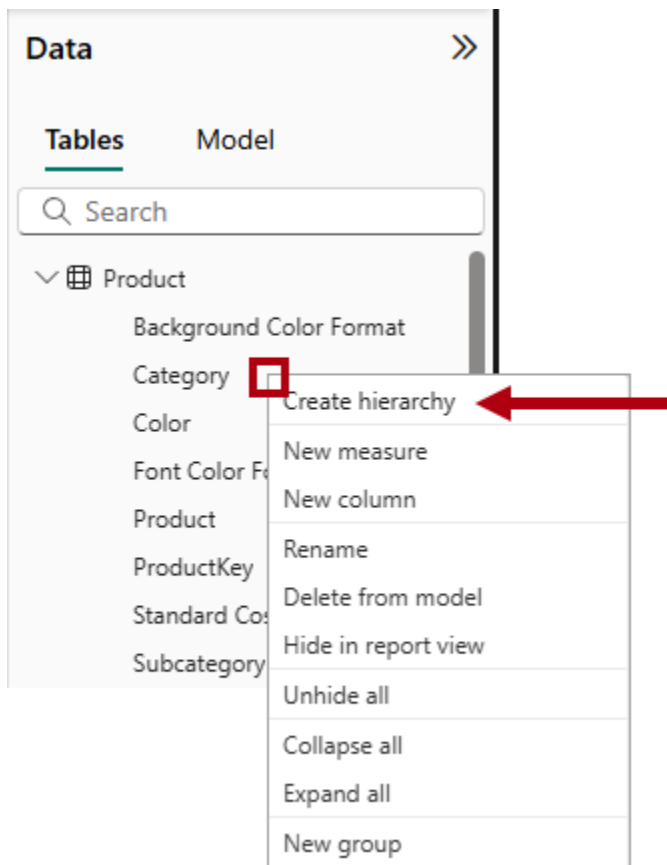
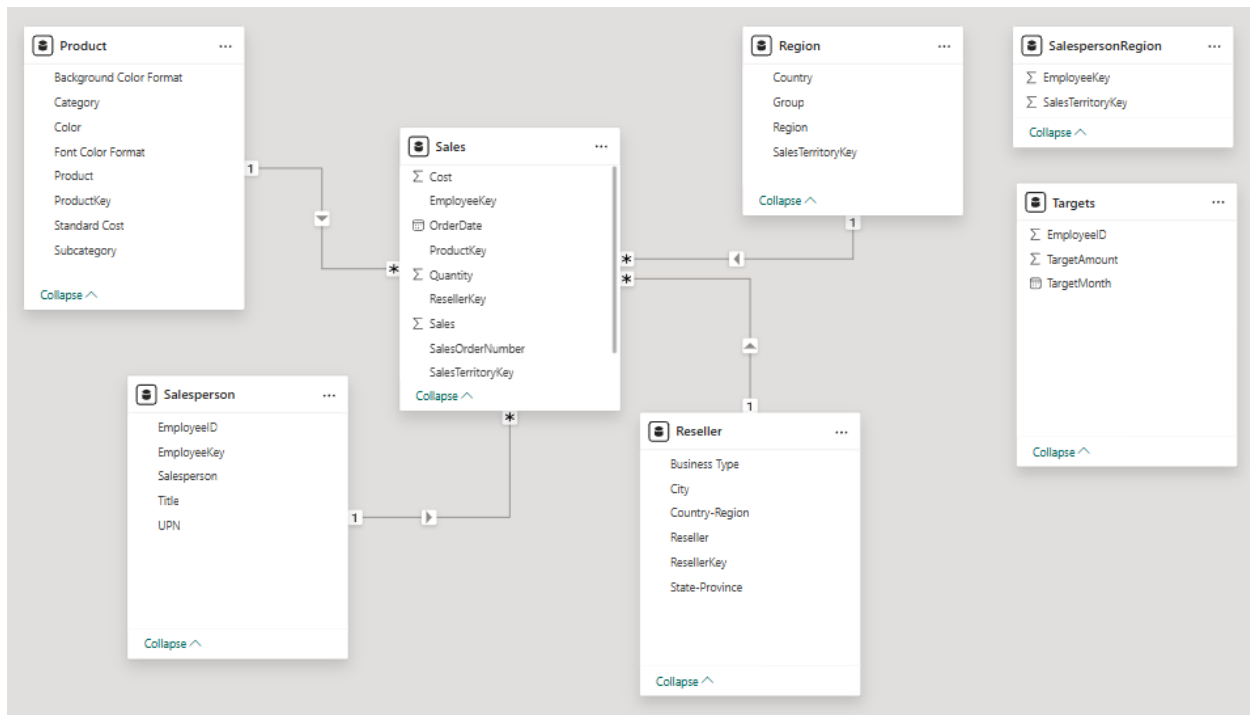
Category	Sum of Sales
Accessories	77,548,570.20
Bikes	77,548,570.20
Clothing	77,548,570.20
Components	77,548,570.20
Total	77,548,570.20

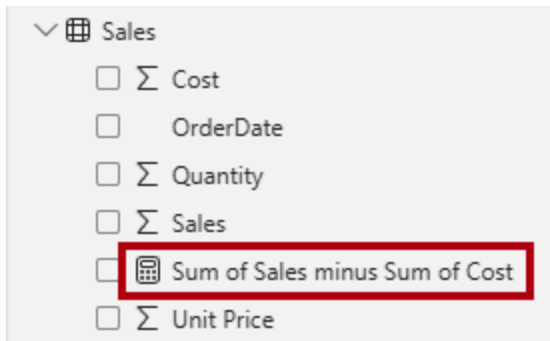
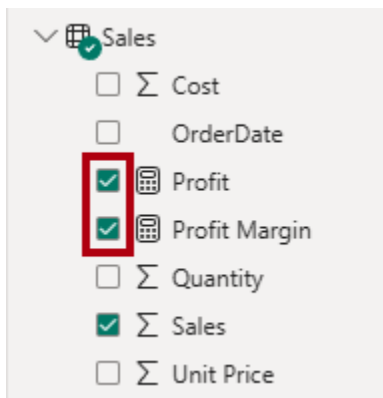
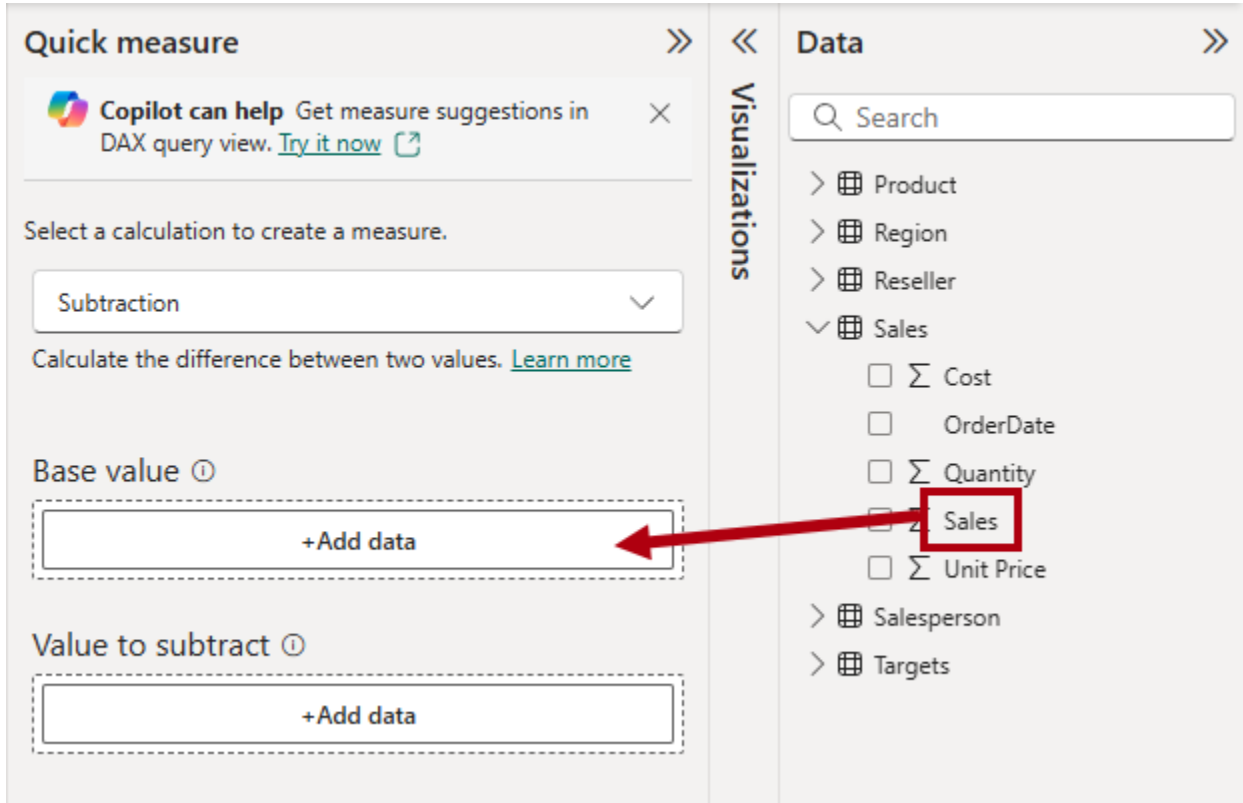
The issue is that the table is based on fields from different tables. The expectation is that each product category displays the sales for that category. However, because there isn't a model relationship between these tables, the Sales table isn't filtered. You'll now add a relationship to propagate filters between the tables.

[PL-300-Microsoft-Power-BI-Data-Analyst/Instructions/Labs/03-configure-semantic-model.md at Main · MicrosoftLearning/PL-300-Microsoft-Power-BI-Data-Analyst](#)

- Create model relationships.
- Configure table and column properties.
- Create hierarchies.
- Create quick measures.
- Configure a many-to-many relationship.

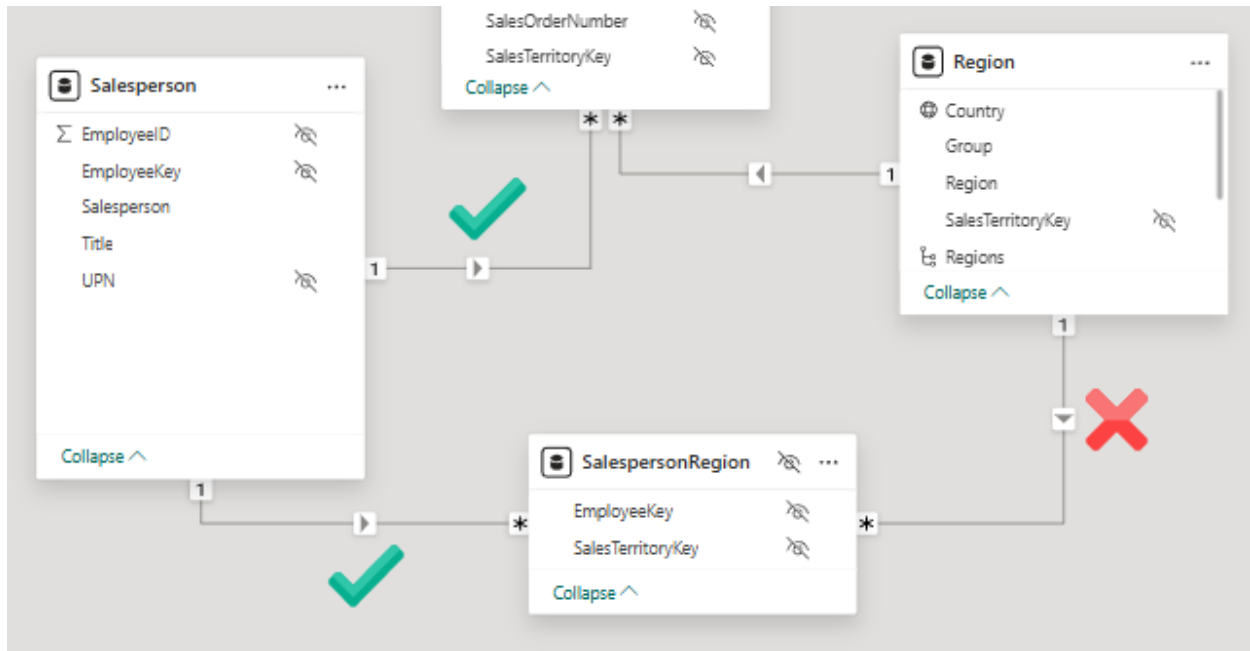
Star Schema





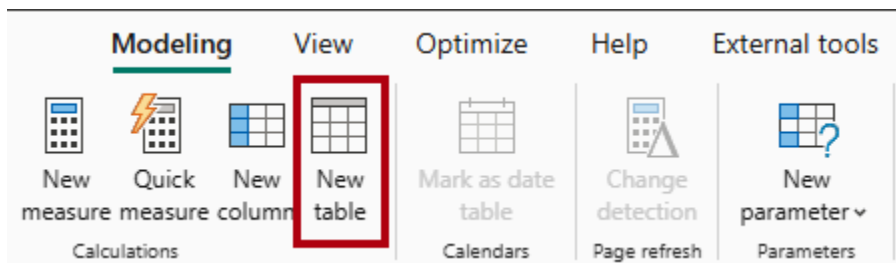
Create a many-to-many relationship between the Salesperson table and the Sales table.

The SalespersonRegion table can be considered to be a bridging table.



[PL-300-Microsoft-Power-BI-Data-Analyst/Instructions/Labs/04-create-dax-calculations.md at Main · MicrosoftLearning/PL-300-Microsoft-Power-BI-Data-Analyst](#)

- Create calculated tables.
- Create calculated columns.
- Create measures.



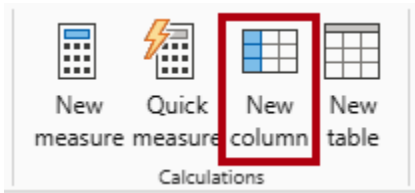
[PL-300-Microsoft-Power-BI-Data-Analyst/Instructions/Labs/04-create-dax-calculations.md at Main · MicrosoftLearning/PL-300-Microsoft-Power-BI-Data-Analyst](#)

Create the Date table

Date =

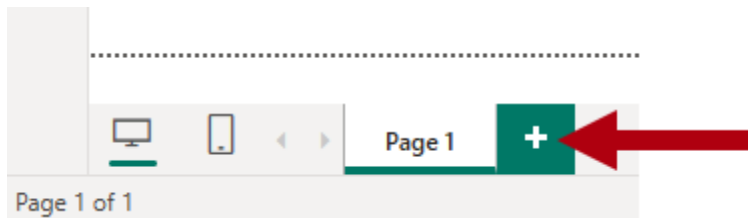
CALENDARAUTO(6)

The CALENDARAUTO function returns a single-column table comprising date values. The "auto" behavior scans all data model date columns to determine the earliest and latest date values stored in the data model. It then creates one row for each date within this range, extending the range in either direction to ensure full years of data is stored.



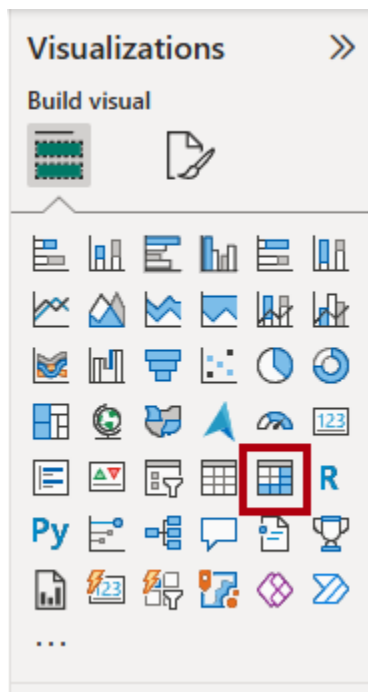
Matrix Visual/Hierarchies

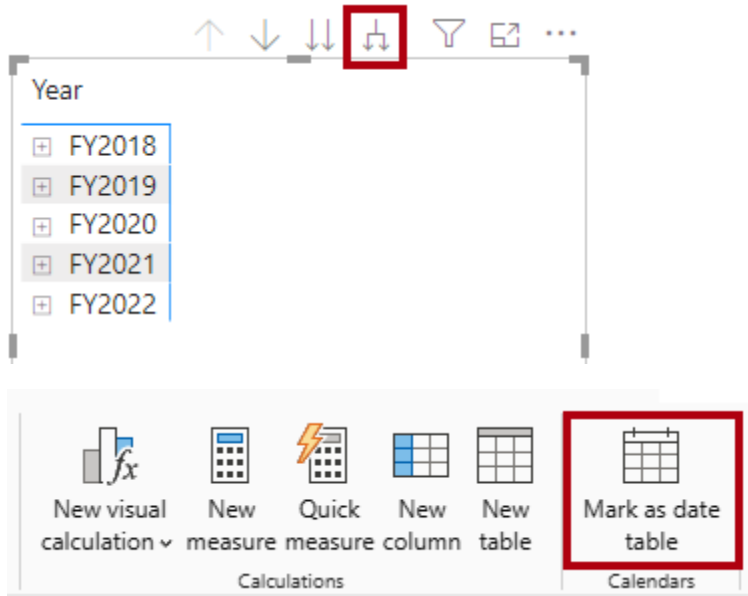
1. To validate the calculations, switch to **Report view**.
2. To create a new report page, select the plus icon next to **Page 1**.



3. To add a matrix visual to the new report page, in the **Visualizations** pane, select the matrix visual type.

Tip: You can hover the cursor over each icon to reveal a tooltip describing the visual type.





Create measures

[PL-300-Microsoft-Power-BI-Data-Analyst/Instructions/Labs/05-modify-dax-filter-context.md](https://www.mediadreamsLLC.com/PL-300-Microsoft-Power-BI-Data-Analyst/Instructions/Labs/05-modify-dax-filter-context.md) at Main · MicrosoftLearning/PL-300-Microsoft-Power-BI-Data-Analyst

- Use the CALCULATE function to manipulate filter context.

create several measures with DAX expressions that use the CALCULATE function to manipulate filter context.

The CALCULATE function is a powerful function you can use to manipulate the filter context. The first argument takes an expression or a measure (a measure is just a named expression). Subsequent arguments allow modifying the filter context.

Add the Sales All Region measure to the matrix visual.

Group	Country	Region	Sum of Sales	Sales All Region
Europe	France	France	\$4,527,840	\$77,548,570.2
		Total	\$4,527,840	\$77,548,570.2
	Germany	Germany	\$1,877,743	\$77,548,570.2
		Total	\$1,877,743	\$77,548,570.2
	United Kingdom	United Kingdom	\$3,883,043	\$77,548,570.2
		Total	\$3,883,043	\$77,548,570.2
	Total		\$10,288,626	\$77,548,570.2
North America	Canada	Canada	\$13,875,633	\$77,548,570.2
		Total	\$13,875,633	\$77,548,570.2
	United States	Central	\$7,633,387	\$77,548,570.2
		Northeast	\$6,715,354	\$77,548,570.2
		Northwest	\$12,004,822	\$77,548,570.2
		Southeast	\$7,638,607	\$77,548,570.2
		Southwest	\$18,001,116	\$77,548,570.2
		Total	\$51,993,286	\$77,548,570.2
	Total		\$65,868,919	\$77,548,570.2
	Pacific	Australia	Australia	\$1,391,025
Total			\$1,391,025	\$77,548,570.2
Total		\$1,391,025	\$77,548,570.2	
Total			\$77,548,570	\$77,548,570.2

[PL-300-Microsoft-Power-BI-Data-Analyst/Instructions/Labs/06-use-dax-time-intelligence.md at Main · MicrosoftLearning/PL-300-Microsoft-Power-BI-Data-Analyst](#)

- Use various time intelligence functions to manipulate filter context that specific concerns dates.

Create a sales year-to-date (YTD) measure by using time intelligence functions.

create a sales YoY growth measure by using a variable.

[PL-300-Microsoft-Power-BI-Data-Analyst/Instructions/Labs/07-create-visual-calculations.md at Main · MicrosoftLearning/PL-300-Microsoft-Power-BI-Data-Analyst](#)

- Create and edit visual calculations.
- Use PREVIOUS(), RUNNINGSUM() and MOVINGAVERAGE() functions to create comparison metrics between each fiscal year.
- Use the optional Axis parameter when creating comparison metrics.
- Use the optional Reset parameter to customize cumulative calculations in a multi-leveled axis.

PBI Service

[PL-300-Microsoft-Power-BI-Data-Analyst/Instructions/Labs/08-design-power-bi-reports.md at Main · MicrosoftLearning/PL-300-Microsoft-Power-BI-Data-Analyst](#)

- Configure visual fields and format properties.
- Sync slicers.
- Publish the report to the Power BI service.

[PL-300-Microsoft-Power-BI-Data-Analyst/Instructions/Labs/09-enhance-power-bi-reports.md at Main · MicrosoftLearning/PL-300-Microsoft-Power-BI-Data-Analyst](#)

- Create a drill through page.
- Apply conditional formatting.
- Create and use bookmarks and buttons.

[PL-300-Microsoft-Power-BI-Data-Analyst/Instructions/Labs/10-perform-analytics-power-bi.md at Main · MicrosoftLearning/PL-300-Microsoft-Power-BI-Data-Analyst](#)

- Create animated scatter charts.
- Use a visual to forecast values.

Row Level Security

[PL-300-Microsoft-Power-BI-Data-Analyst/Instructions/Labs/11-secure-data-access.md at Main · MicrosoftLearning/PL-300-Microsoft-Power-BI-Data-Analyst](#)

- Implement dynamic row-level security (RLS) in Power BI.
- Create and test a role using USERPRINCIPALNAME().

*a post-publication task to map security principals to the **Salespeople** role*

after you've defined the DAX RLS role in Power BI Desktop and published the model, tailored to the PL-300 Lab 11 flow.

Post-publication steps: Map security principals to a DAX RLS role

1. **Open the Power BI Service**
 - Go to the workspace where the dataset (semantic model) was published.
2. **Open dataset (semantic model) security**
 - Select **More options (...)** → **Security** on the dataset.
3. **Locate the RLS role**
 - Find the role you created in Desktop (for example: SalesTerritoryRole).
4. **Add security principals**
 - Add one or more:
 - Azure AD users
 - Security groups
 - Microsoft Entra ID (AAD) groups
 -  Best practice: use **AAD security groups**, not individual users.
5. **Confirm role membership**
 - Ensure each principal appears under the correct role.
 - Save changes.
6. **Test RLS in the Service**
 - Use **"Test as role"** or **"View as"** to validate filtered results.
7. **Assign report access**
 - Make sure users/groups have:

- At least **Viewer** access to the workspace **or**
- Shared access to the report/app
- RLS does **not** grant access by itself.

8. (Optional) App validation

- If using a Power BI App, test access via the App experience to ensure RLS is honored.

One-line summary

Publish → Dataset Security → Add users/groups to role → Save → Test → Grant report access

Key Ways to "Lock" Power BI Reports

- **Lock Visuals (Layout):** Go to the **View** tab and check "**Lock objects**" to lock all visuals on the page, preventing accidental movement or resizing.
- **Prevent Editing (Service):** Publish the report to the Power BI Service, then share via an App with "Viewer" role permissions to restrict editing capabilities.
- **Lock Specific Filters:** Within the Filters pane, you can lock specific filters so end-users cannot change them, as explained in.
- **Prevent Data Access:** Use Row-Level Security (RLS) to restrict the data users can see.

Limitations

- **Lock objects** only locks position in the desktop editor; it does not stop users with edit access to the report from turning this setting off.
- To prevent users from editing or moving objects entirely, they should only be given viewing access to the published report, not editing rights.

Hey Co - How do you lock an entire PBI report so the end user can only view it with PBI Desktop?

Copilot said:

Copilot

Short answer: **You can't fully "lock" a Power BI report so it only opens in Desktop**, but you *can* prevent end users from opening it anywhere *except* Desktop **by controlling distribution and not publishing it to the Power BI Service.**

Here's the full breakdown and the closest possible approaches:

What Power BI *does* allow

Power BI gives you control over:

1. Who can access a published report

- Using the Power BI Service, you can restrict viewing permissions.
- But this doesn't stop someone from viewing it **in the Service**, which is exactly what you *don't* want.

2. Who can open the PBIX file

You can distribute the .pbix file directly.

If you never publish it to the Power BI Service, **Desktop is the only way to open it.**

 **This is the only real way to enforce Desktop-only viewing.**

What Power BI *does NOT* support

Power BI Desktop **does NOT** support:

- Password-protecting a PBIX file
- Locking visuals or pages behind a password
- Restricting viewing mode (e.g., forcing "View Only" inside Desktop)
- Encryption keys at the PBIX level beyond what Windows provides
- Publishing a report but blocking users from opening it online

So you cannot make a "view-only PBIX" or "Desktop-only PBIX" through any built-in setting.

 **Practical methods organizations use to enforce Desktop-only viewing**

Method 1 — Never publish to the Power BI Service

If you simply never publish:

- Users must open the PBIX in Power BI Desktop.
- You keep control by distributing PBIX via:
 - SharePoint/OneDrive with **read-only** permissions
 - A shared network location
 - A version-controlled repository

This is the standard approach.

Method 2 — Give users “Read” access to the PBIX file

This prevents them from editing or copying it.

On SharePoint or OneDrive:

- Set permissions to **View Only**
- Disable **Download** (if using SharePoint)
- That way, they can open the PBIX from the cloud but cannot overwrite or republish it.

Method 3 — Use AutoRecover/Recoverable File Restrictions

Not a security measure, but you can reduce risk by:

- Turning off AutoSave
- Storing PBIX files in read-only folders

Method 4 — If the worry is data exposure, use RLS

Row-Level Security applies even in Desktop when the user signs in.

Useful if you're distributing PBIX files but want data-level control.

If your goal is to prevent editing

There is no “lock editing” feature, but you can do this:

Use Power BI Desktop Report View mode

Tell users:  *Don't switch to Model View*

 *Don't switch to Data View*

These require the original model. If they don't have write access to the file location, they can't save changes.

But this depends on user behavior—not a true lock.

Summary: The closest solution

Goal	Best Available Approach
Only view in Desktop	Do not publish to Power BI Service; distribute PBIX only.

Goal	Best Available Approach
Prevent editing	Store PBIX in read-only location; give users read access only.
Secure data	Use RLS + read-only file distribution.